Omar Ahmed Ghareeb

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LinkedIn | GitHub | Portfolio

Professional Profile

Media Engineering Technology graduate from the German University in Cairo (GUC) with hands-on experience in software development, AI applications, full-stack web development, mobile development, and interactive systems. Proficient in Java, Python, C/C++, JavaScript, Dart, SQL, and C, with practical experience building full-stack applications, AI-powered mobile solutions, database systems, and 2D/3D games. Strong foundation in object-oriented programming, software engineering, data processing, machine learning, and system design, with a proven ability to learn quickly, solve complex problems, and work effectively in collaborative development environments.

Education

German University in Cairo (GUC) Cairo, Egypt
B.Sc. in Engineering (Media Engineering Technology) *2026*

Technical Skills

Programming Languages	:	Java, Python, C++, C, JavaScript, Dart, SQL, C#
Web & Mobile	:	HTML, CSS, React, Next.js, Tailwind CSS, Flutter, Node.js
AI & Data	:	TensorFlow Lite, NumPy, Machine Learning, Data Processing, Data Visualization
Tools & Concepts	:	Git, Visual Studio Code, Unity 3D, REST APIs, Microservices, Web Scraping, Arduino, Power BI, Object-Oriented Programming, Database Design

Experience

Software Developer (Intern) Jul 2024 – Sep 2024
Sportyano for Electronic and Digital Systems *Giza, Egypt*

- Contributed to software projects in the sports technology domain within the IT department.
- Assisted in application design and development; wrote and tested code; supported project planning and execution.
- Participated in code reviews, debugging, and troubleshooting; strengthened proficiency in Python and Java.
- Built teamwork, problem-solving, and professional communication skills in a real-world development environment.

Projects

Specto – AI-Powered Mobile Audio Analyzer *Flutter, Dart, TensorFlow Lite, DSP*

- Engineered a cross-platform mobile application for real-time audio acquisition, processing, visualization, and AI-powered sound classification.
- Designed and implemented a complete **Digital Signal Processing (DSP)** pipeline using **FFT**, **STFT**, Hann Windowing, and RMS energy analysis for live microphone input.
- Developed interactive real-time visualizations including waveform, frequency spectrum, spectrogram, dominant frequency, and audio energy monitoring.
- Integrated **TensorFlow Lite** to perform on-device AI sound classification, enabling low-latency offline inference while preserving user privacy.
- Optimized audio buffering, memory management, and rendering to deliver smooth real-time performance on Android devices.

Attack on Titans: Utopia & The Last of Us: Legacy *Java*

- Developed two Java-based games from scratch using object-oriented programming concepts.
- Designed interactive gameplay mechanics, implemented enemy AI, and optimized performance for smooth execution.

Rafiki – Full-Stack Travel Website *HTML, CSS, JavaScript, MySQL*

- Built a responsive travel platform with interactive booking features.
- Designed the frontend using HTML/CSS/JavaScript and integrated backend services with MySQL, enabling seamless data handling and user experience.

3D Graphical Game *C*

- Created a Temple Run-inspired 3D graphical game using C.
- Implemented rendering pipelines, collision detection, and animation logic to deliver real-time interactive gameplay.

Unity 3D Games *Unity, C#*

- Designed and developed two Unity-based games: an endless runner and a Diablo-style action RPG.
- Focused on level design, user interaction, and gameplay balancing while applying Unity physics and asset integration.

Internship Management System *Database Project*

- Worked on designing and implementing a relational database for managing student internship applications.
- Defined functional requirements, created ER diagrams, and enforced business rules such as blacklisted companies, application cycles, and internship validation.

Processor Design and Simulation *Computer Architecture Project (CSEN601)*

- Developed a C-based processor simulator, modeling Von Neumann and Harvard architectures.
- Implemented instruction sets, pipelining, and branch/jump handling, while analyzing clock cycles and hazards.

Certifications

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional Issued by Oracle
October 2025 *Expires: October 31, 2027*

- Earned the **OCI Generative AI Professional** certification, demonstrating proficiency in modern AI application development on Oracle Cloud.
- Strong understanding of **Large Language Models (LLMs)**, including training, evaluation, prompting, and deployment workflows.
- Experienced with **Retrieval-Augmented Generation (RAG)**, including embedding models, semantic search, and vector databases.
- Familiar with **LangChain**, enabling the orchestration, tracing, and evaluation of end-to-end LLM applications.

Languages

- Arabic: Native or Bilingual Proficiency
- English: Full Professional Proficiency
- German: A2 Certified